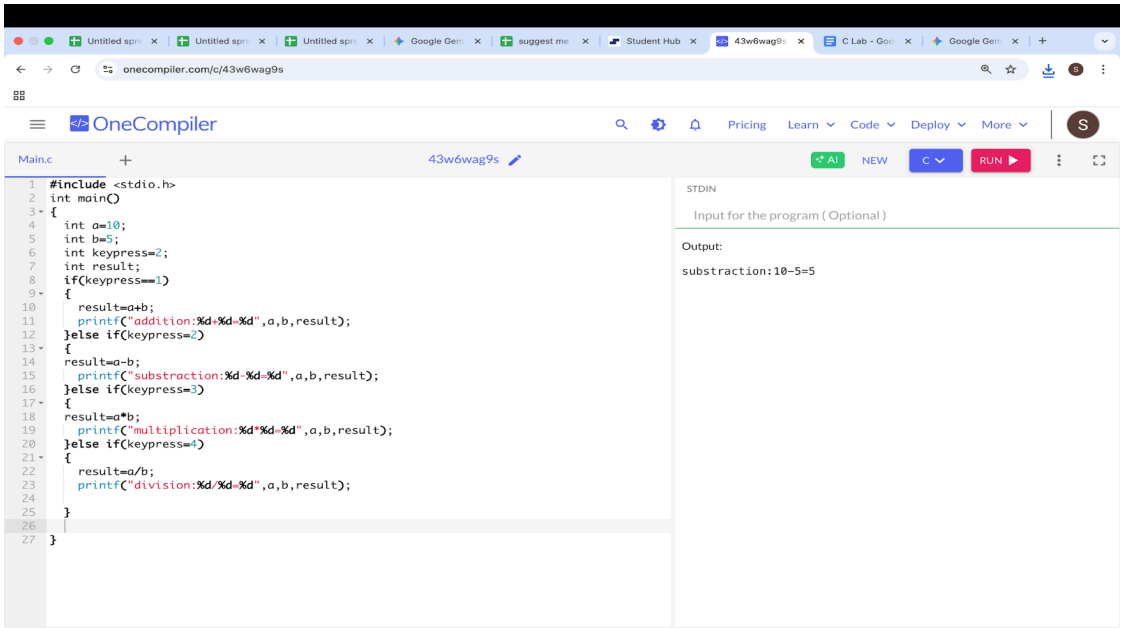
1) DESIGNING SIMPLE CALCULATOR

```
#include <stdio.h>
int main()
{
   int a=10;
   int b=5;
   int keypress=2;
   int result;
   if(keypress==1)
   {
      result=a+b;
      printf("addition:%d-%d:%d",a,b,result);
   }
   else if(keypress==2)
   {
      result=a-b;
      printf("substraction:%d-%d:%d",a,b,result);
   }
   else if(keypress==3)
   {
      result=a*b;
      printf("multiplication:%d-%d:%d",a,b,result);
   }
   else if(keypress==4)
   {
      result=a/b;
      printf("division:%d-%d:%d",a,b,result);
   }
}
```



2) Converting fahrenheit into degree celcius

```
#include <stdio.h>
int main()
 {
   float fahrenheit;
   float celsius;

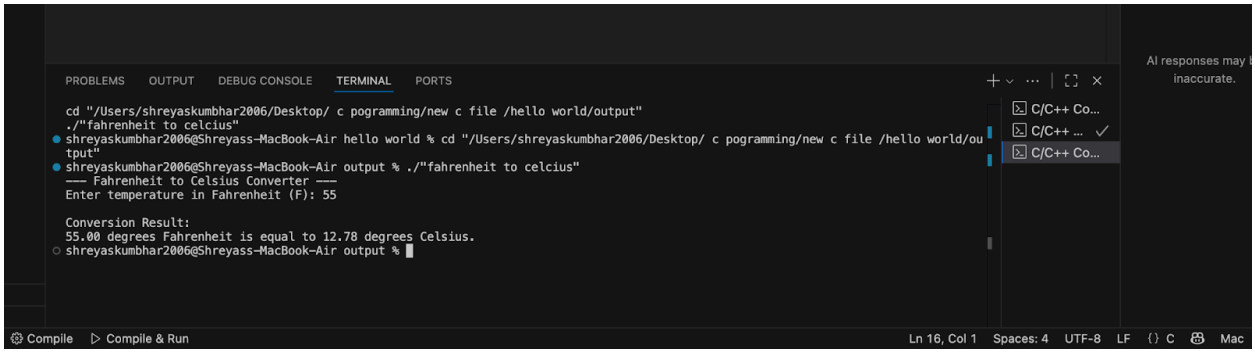
   printf("--- Fahrenheit to Celsius Converter ---\n");
   printf("Enter temperature in Fahrenheit (F): ");

   if (scanf("%f", &fahrenheit) != 1)
  {
      printf("Error: Invalid input. Please enter a numerical value.\n");
      return 1;
   }

   celsius = (fahrenheit - 32.0) * (5.0 / 9.0);

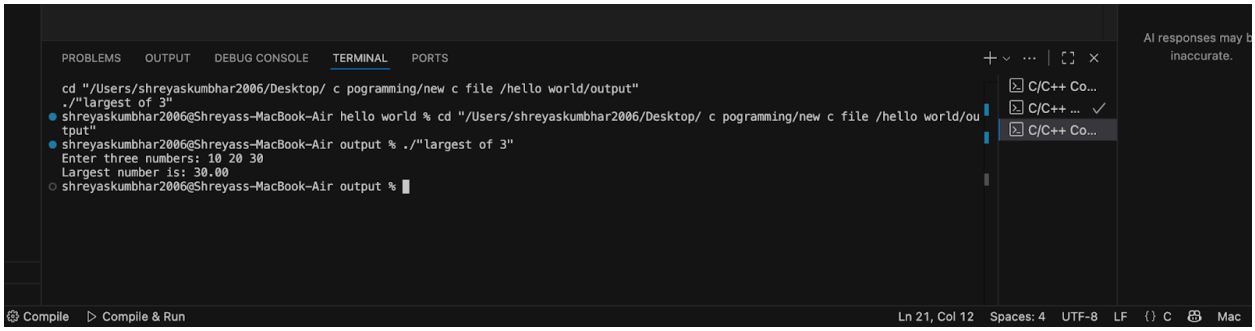
   printf("\nConversion Result:\n");
   printf("%.2f degrees Fahrenheit is equal to %.2f degrees Celsius.\n", fahrenheit,
celsius);

   return 0;
}
```



3) Determining largest among three numbers

```
#include <stdio.h>
int main()
 {
   float a, b, c;
   printf("Enter three numbers: ");
   scanf("%f %f %f", &a, &b, &c);
   if (a >= b && a >= c)
{
      printf("Largest number is: %.2f\n", a);
   } else if (b >= a && b >= c)
{
      printf("Largest number is: %.2f\n", b);
   } else
{
      printf("Largest number is: %.2f\n", c);
   }
   return 0;
}
```



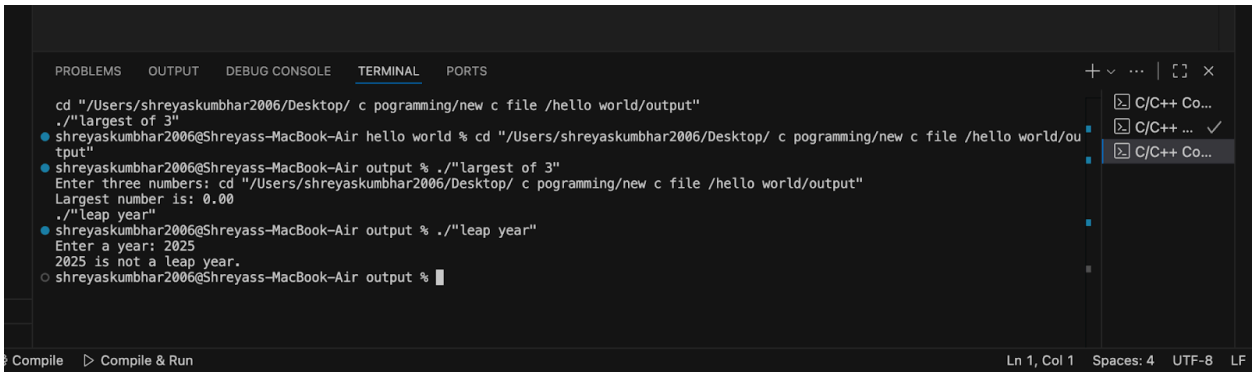
4)Determining leap year

```c
#include <stdio.h>
int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if (year % 400 == 0) {
        printf("%d is a leap year.\n", year);
    } else if (year % 100 == 0) {
        printf("%d is not a leap year.\n", year);
    } else if (year % 4 == 0) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```



5)Eligibility for admission

```
#include <stdio.h>
int main() {
   int physics, maths, chemistry;

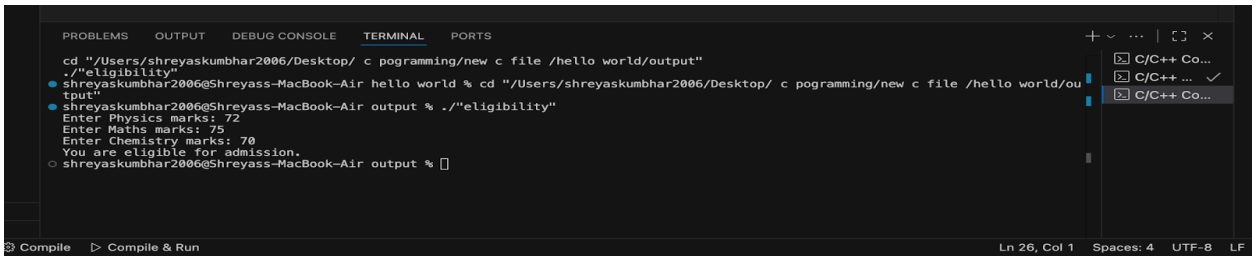
   printf("Enter Physics marks: ");
   scanf("%d", &physics);

   printf("Enter Maths marks: ");
   scanf("%d", &maths);

   printf("Enter Chemistry marks: ");
   scanf("%d", &chemistry);

   if (physics >= 65 && maths >= 65 && chemistry >= 65) {
      printf("You are eligible for admission.\n");
   } else {
      printf("You are NOT eligible for admission.\n");
   }

   return 0;
}
```



6) Determining in which quadrant point lies

```
#include <stdio.h>
int main()
 {
   float x, y;
   printf("Enter x coordinate: ");
   scanf("%f", &x);
   printf("Enter y coordinate: ");
   scanf("%f", &y);
   if (x > 0 && y > 0) {
      printf("Point lies in Quadrant I\n");
   } else if (x < 0 && y > 0) {
      printf("Point lies in Quadrant II\n");
   } else if (x < 0 && y < 0) {
      printf("Point lies in Quadrant III\n");
   } else if (x > 0 && y < 0) {
      printf("Point lies in Quadrant IV\n");
   } else if (x == 0 && y == 0) {
      printf("Point lies at the Origin\n");
   } else if (x == 0) {
      printf("Point lies on the Y-axis\n");
   } else if (y == 0) {
      printf("Point lies on the X-axis\n");
   }
}
```



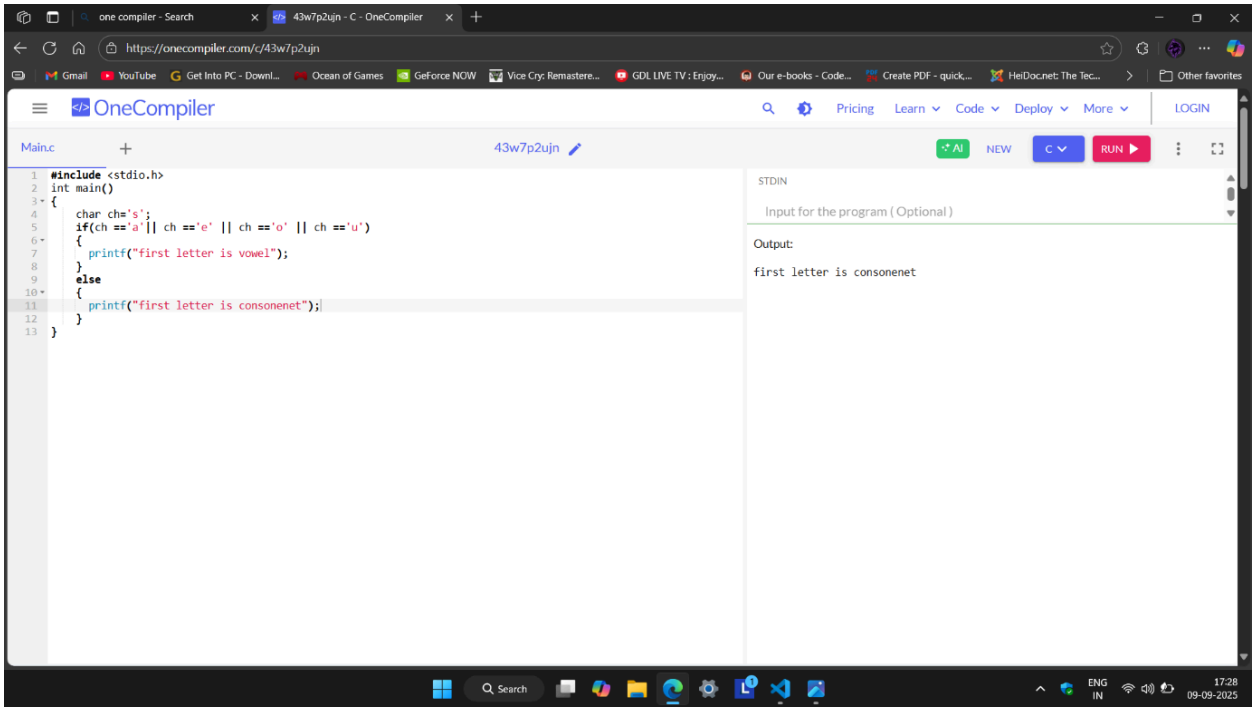
7) Determining the alphabet is vowel or consonant

```
#include <stdio.h>

int main() {
   char ch = 's';

   if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
      printf("First letter is vowel");
   } else {
      printf("First letter is consonant");
   }

   return 0;
}
```



8) Designing calculator using switch case

```
#include <stdio.h>
int main() {
   float num1, num2;
   char op;
   printf("Enter first number: ");
   scanf("%f", &num1);
   printf("Enter an operator (+, -, *, /): ");
   scanf(" %c", &op);   // space added so it reads operator correctly
   printf("Enter second number: ");
   scanf("%f", &num2);
   switch (op) {
      case '+':
         printf("Result: %.2f\n", num1 + num2);
         break;
      case '-':
         printf("Result: %.2f\n", num1 - num2);
         break;
      case '*':
         printf("Result: %.2f\n", num1 * num2);
         break;
      case '/':
         if (num2 != 0) {
            printf("Result: %.2f\n", num1 / num2);
         } else {
            printf("Error: Division by zero not allowed.\n");
         }    break;
```

```
        default:
            printf("Invalid operator.\n");
    }

    return 0;
}
```



9) Calculator using switch case

```
#include <stdio.h>
int main() {
    float num1, num2;
    char op;

    printf("Enter first number: ");
    scanf("%f", &num1);

    printf("Enter an operator (+, -, *, /): ");
    scanf(" %c", &op);   // space added so it reads operator correctly

    printf("Enter second number: ");
    scanf("%f", &num2);

    switch (op) {
        case '+':
            printf("Result: %.2f\n", num1 + num2);
            break;

        case '-':
            printf("Result: %.2f\n", num1 - num2);
            break;

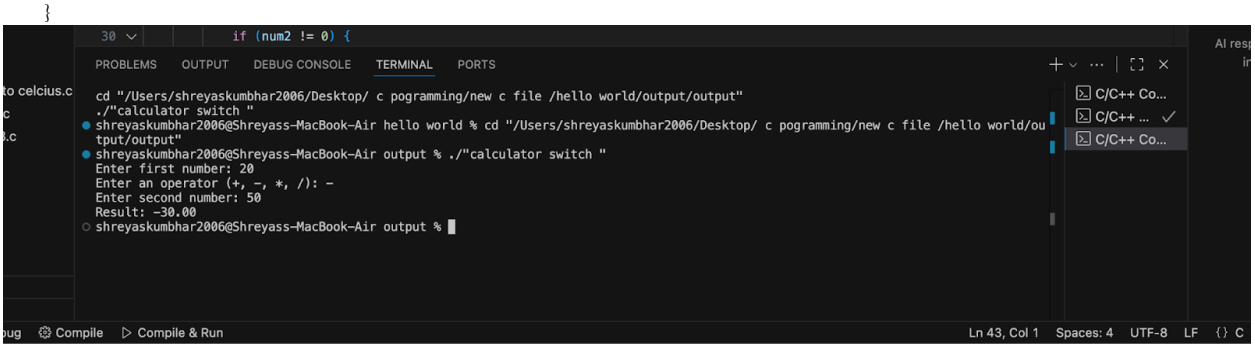
        case '*':
            printf("Result: %.2f\n", num1 * num2);
            break;

        case '/':
            if (num2 != 0) {
                printf("Result: %.2f\n", num1 / num2);
            } else {
                printf("Error: Division by zero not allowed.\n");
            }

            break;

        default:
            printf("Invalid operator.\n");
    }

    return 0;
}
```



10) Electricity bill calculator

```
#include <stdio.h>

int main() {
    int units;
    double bill_amount = 0.0;
    double surcharge_rate = 0.20;

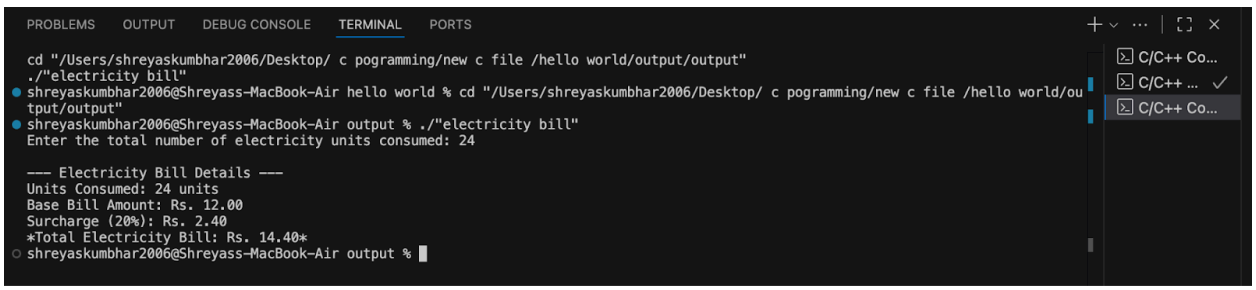
    printf("Enter the total number of electricity units consumed: ");
    if (scanf("%d", &units) != 1 || units < 0) {
        printf("Invalid input. Please enter a non-negative integer for units.\n");
        return 1;
    }

    if (units <= 50) {
        bill_amount = units * 0.50;
    } else if (units <= 150) {
        bill_amount = 50 * 0.50;
        bill_amount += (units - 50) * 0.75;
    } else if (units <= 250) {
        bill_amount = 50 * 0.50;
        bill_amount += 100 * 0.75;
        bill_amount += (units - 150) * 1.20;
    } else {
        bill_amount = 50 * 0.50;
        bill_amount += 100 * 0.75;
        bill_amount += 100 * 1.20;
        bill_amount += (units - 250) * 1.50;
    }

    double base_bill = bill_amount;
    double surcharge_amount = base_bill * surcharge_rate;
    bill_amount += surcharge_amount;

    printf("\n--- Electricity Bill Details ---\n");
    printf("Units Consumed: %d units\n", units);
    printf("Base Bill Amount: Rs. %.2lf\n", base_bill);
    printf("Surcharge (20%%): Rs. %.2lf\n", surcharge_amount);
    printf("*Total Electricity Bill: Rs. %.2lf*\n", bill_amount);

    return 0;
}
```



11) Calculating simple interest

```
#include <stdio.h>

int main() {
   float principal, rate, time, si;

   printf("Enter Principal Amount: ");
   scanf("%f", &principal);

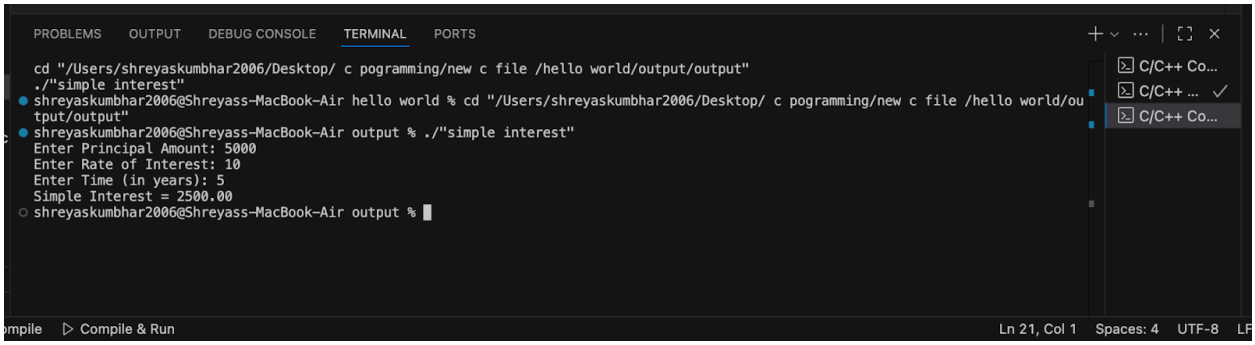
   printf("Enter Rate of Interest: ");
   scanf("%f", &rate);

   printf("Enter Time (in years): ");
   scanf("%f", &time);

   si = (principal * rate * time) / 100;

   printf("Simple Interest = %.2f\n", si);

   return 0;
}
```



12) Printing numbers using for loop

```
#include <stdio.h>

int main() {
   int i;

   for (i = 1; i <= 10; i++) {
      printf("%d\n", i);
   }

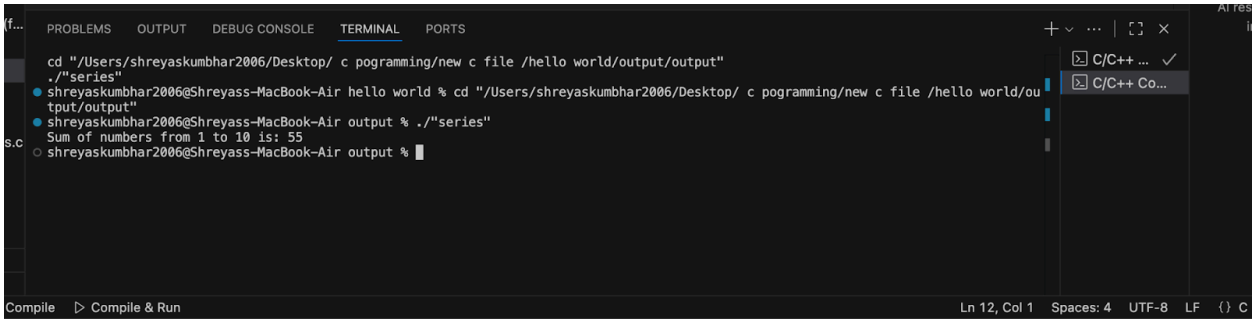
   return 0;
}
```



13) Sum of series using for loop

```
#include <stdio.h>

int main() {
   int sum = 0;
   int i;
   for(i = 1; i <= 10; i++) {
      sum += i;
   }
   printf("Sum of numbers from 1 to 10 is: %d\n", sum);
   return 0;
}
```



14)Fibonacci series using for loop

```c
#include <stdio.h>

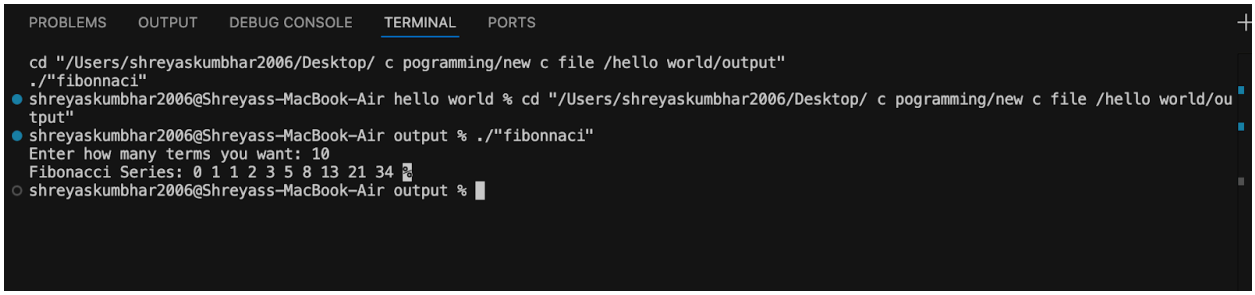
int main() {
   int n, first = 0, second = 1, next;

   printf("Enter how many terms you want: ");
   scanf("%d", &n);

   printf("Fibonacci Series: ");

   for (int i = 1; i <= n; i++) {
      printf("%d ", first);
      next = first + second;
      first = second;
      second = next;
   }

   return 0;
}
```



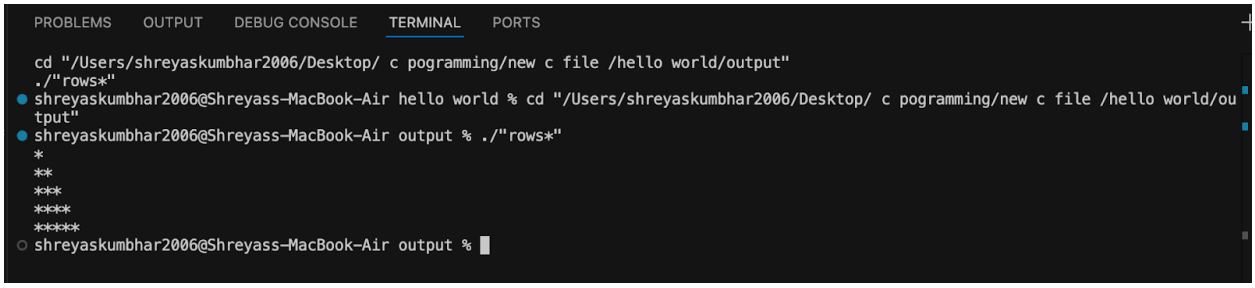
15) Printing pattern using for loop (Rows of *)

```
#include <stdio.h>

int main() {
   int i, j;
   int n = 5;  // Number of rows for the triangle

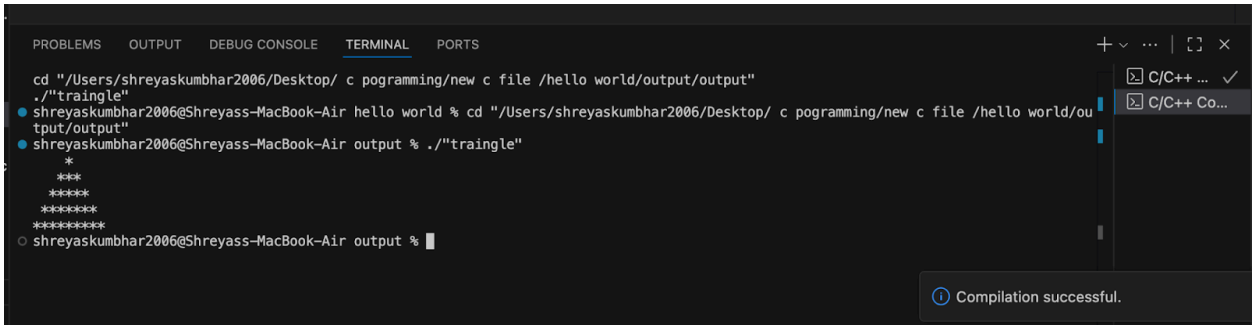
   for(i = 1; i <= n; i++) {
      for(j = 1; j <= i; j++) {
         printf("*");
      }
      printf("\n");
   }

   return 0;
}
```



16) Printing pattern using for loop (triangle of *)

```
#include <stdio.h>
int main() {
   int i, j, k;
   int n = 5;
   for(i = 1; i <= n; i++) {
      for(j = i; j < n; j++) {
         printf(" ");
      }
      for(k = 1; k <= (2*i - 1); k++) {
         printf("*");
      }
      printf("\n");
   }
}
```



17) Printing pattern using for loop

```
#include <stdio.h>

int main() {
    int physics, chemistry, biology, mathematics, computer;
    int total_marks;
    float percentage;

    scanf("%d %d %d %d %d", &physics, &chemistry, &biology, &mathematics, &computer);

    total_marks = physics + chemistry + biology + mathematics + computer;

    percentage = (float)total_marks / 5.0;

    printf("Percentage: %.2f\n", percentage);

    if (percentage >= 90.0) {
        printf("Grade A\n");
    } else if (percentage >= 80.0) {
        printf("Grade B\n");
    } else if (percentage >= 70.0) {
        printf("Grade C\n");
    } else if (percentage >= 60.0) {
        printf("Grade D\n");
    } else if (percentage >= 40.0) {
        printf("Grade E\n");
    } else {
        printf("Grade F\n");
    }

    return 0;
}
```



18)Eligibility for admission using functions

```
#include <stdio.h>

float calculatePercentage(int p, int c, int b, int m, int comp, int max_marks);
void findGrade(float percentage);

int main() {
   int physics, chemistry, biology, mathematics, computer;
   float percentage;
   int max_marks_per_subject = 100;
   int total_max_marks = 5 * max_marks_per_subject;

   printf("Enter marks for five subjects (out of 100 each):\n");
   printf("Physics: ");
   scanf("%d", &physics);
   printf("Chemistry: ");
   scanf("%d", &chemistry);
   printf("Biology: ");
   scanf("%d", &biology);
   printf("Mathematics: ");
   scanf("%d", &mathematics);
   printf("Computer: ");
   scanf("%d", &computer);

   percentage = calculatePercentage(physics, chemistry, biology, mathematics, computer, total_max_marks);

   printf("\nTotal Marks Obtained: %d\n", (int)(percentage * total_max_marks / 100.0));
   printf("Total Percentage: %.2f%%\n", percentage);

   findGrade(percentage);

   return 0;
}

float calculatePercentage(int p, int c, int b, int m, int comp, int max_marks) {
   int total_marks = p + c + b + m + comp;
   float per = ((float)total_marks / max_marks) * 100.0;

   return per;
}

void findGrade(float percentage) {
   printf("Calculated Grade: ");

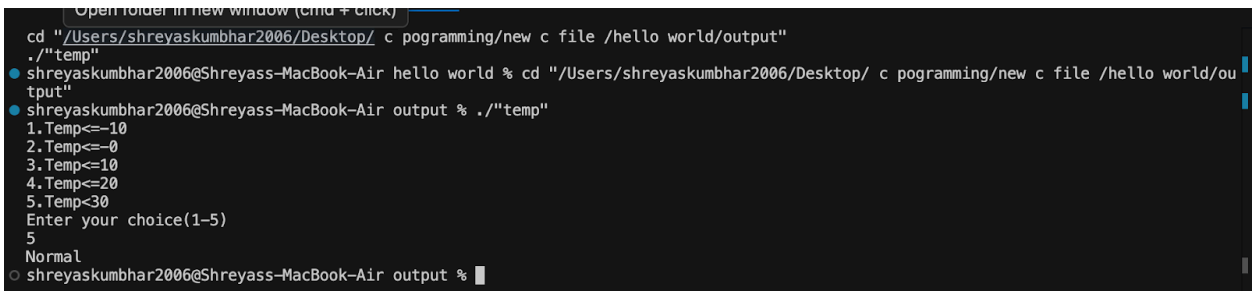

   if (percentage >= 90) {
      printf("Grade A\n");
   } else if (percentage >= 80) {
      printf("Grade B\n");
   } else if (percentage >= 70) {
      printf("Grade C\n");
   } else if (percentage >= 60) {
      printf("Grade D\n");
   } else if (percentage >= 40) {
      printf("Grade E\n");
   } else {
      printf("Grade F\n");
   }
}
```

```
cd "/Users/shreyaskumbhar2006/Desktop/ c pogramming/new c file /hello world/output"
./"eligibility using functions"
shreyaskumbhar2006@Shreyass-MacBook-Air hello world % cd "/Users/shreyaskumbhar2006/Desktop/ c pogramming/new c file /hello world/output"
shreyaskumbhar2006@Shreyass-MacBook-Air output % ./"eligibility using functions"
Enter marks for five subjects (out of 100 each):
Physics: 75
Chemistry: 80
Biology: 90
Mathematics: 65
Computer: 78

Total Marks Obtained: 388
Total Percentage: 77.60%
Calculated Grade: Grade C
```

19)Reading temperature and displaying suitable message

```
#include <stdio.h>
int main()
{
   int choice;
   printf("1.Temp<=-10 \\n");
   printf("2.Temp<=-0 \\n");
   printf("3.Temp<=10 \\n");
   printf("4.Temp<=20 \\n");
   printf("5.Temp<30 \\n");
   printf("Enter your choice(1-5)\\n");
   scanf("%d", &choice);
   switch (choice)
   {
   case 1:
      printf("Deep freeze \\n");
      break;
   case 2:
      printf("Freeze \\n");
      break;
   case 3:
      printf("Cold \\n");
      break;
   case 4:
      printf("Cool \\n");
      break;
   case 5:
      printf("Normal \\n");
      break;
   default:
      printf("Hot \\n");
   }
   return 0;
}
```



20)Determining whether an alphabet is vowel or consonants using functions

```
#include <stdio.h>
#include <ctype.h>   // for tolower()

int checkVowel(char ch) {
    ch = tolower(ch);

    if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u') {
        return 1;   // vowel
    } else {
        return 0;   // consonant
    }
}

int main() {
    char ch;

    printf("Enter any alphabet: ");
    scanf("%c", &ch);

    if( (ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z') ) {

        if (checkVowel(ch)) {
            printf("%c is a vowel\n", ch);
        } else {
            printf("%c is a consonant\n", ch);
        }

    } else {
        printf("Invalid input. Please enter an alphabet.\n");
    }

    return 0;
}
```



21)Electricity bill calculator using functions

```c
#include <stdio.h>

float calculate_bill(int units);

float calculate_bill(int units) {
    float total_cost = 0.0;
    float surcharge = 0.0;

    if (units <= 50) {
        total_cost = units * 0.50;
    } else if (units <= 150) {
        total_cost = 50 * 0.50;
        total_cost += (units - 50) * 0.75;
    } else if (units <= 250) {
        total_cost = 50 * 0.50;
        total_cost += 100 * 0.75;
        total_cost += (units - 150) * 1.20;
    } else {
        total_cost = 50 * 0.50;
        total_cost += 100 * 0.75;
        total_cost += 100 * 1.20;
        total_cost += (units - 250) * 1.50;
    }

    surcharge = total_cost * 0.20;

    return total_cost + surcharge;
}

int main() {
    int consumed_units;
    float final_bill;

    printf("Welcome to the Electricity Bill Calculator\n");
    printf("---------------------------------------\n");

    printf("Enter the total electricity units consumed: ");
    if (scanf("%d", &consumed_units) != 1 || consumed_units < 0) {
        printf("Invalid input. Please enter a positive integer for units.\n");
        return 1;
    }

    final_bill = calculate_bill(consumed_units);

    printf("\n--- Calculation Summary ---\n");
    printf("Units Consumed: %d units\n", consumed_units);
    printf("Total Bill Amount (including 20%% surcharge): Rs. %.2f\n", final_bill);
    printf("---------------------------\n");

    return 0;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

cd "/Users/shreyaskumbhar2006/Desktop/ c pogramming/new c file /hello world/output"
./"electricitybillfunction"
shreyaskumbhar2006@Shreyass-MacBook-Air hello world % cd "/Users/shreyaskumbhar2006/Desktop/ c pogramming/new c file /hello world/output"
shreyaskumbhar2006@Shreyass-MacBook-Air output % ./"electricitybillfunction"
Welcome to the Electricity Bill Calculator
---------------------------------------
Enter the total electricity units consumed: 67

--- Calculation Summary ---
Units Consumed: 67 units
Total Bill Amount (including 20% surcharge): Rs. 45.30
---------------------------
shreyaskumbhar2006@Shreyass-MacBook-Air output %
```

22) Sum of series of number in using function

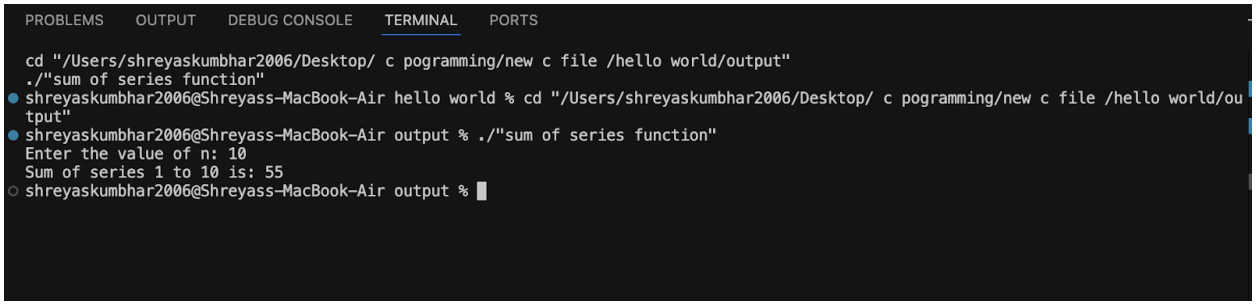
```
#include <stdio.h>

int sumSeries(int n) {
   int sum = 0;
   for(int i = 1; i <= n; i++) {
      sum = sum + i;
   }
   return sum;
}

int main() {
   int n;
   printf("Enter the value of n: ");
   scanf("%d", &n);

   int result = sumSeries(n);
   printf("Sum of series 1 to %d is: %d\n", n, result);

   return 0;
}
```



23) Converting decimal number into binary number using function

```
#include <stdio.h>

long decimalToBinary(int n) {
    long binary = 0;
    long place = 1;
    int rem;

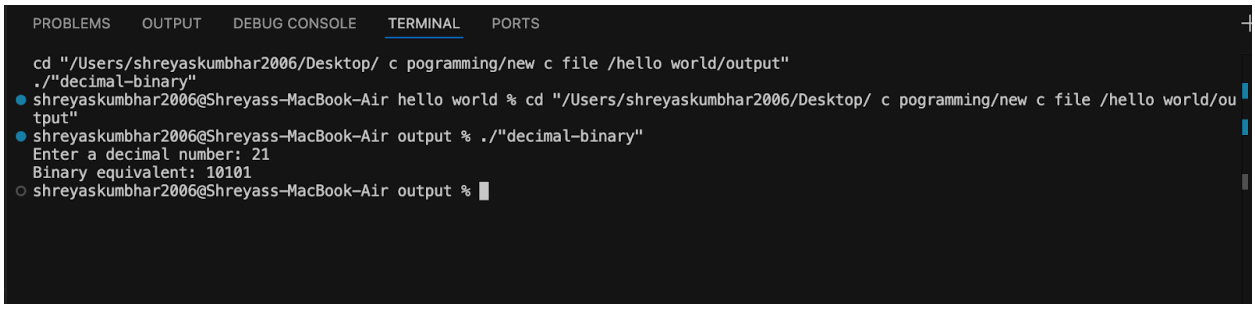
    while(n > 0) {
        rem = n % 2;
        binary = binary + rem * place;
        place = place * 10;
        n = n / 2;
    }

    return binary;
}

int main() {
    int n;
    printf("Enter a decimal number: ");
    scanf("%d", &n);

    long result = decimalToBinary(n);
    printf("Binary equivalent: %ld\n", result);

    return 0;
}
```



24) Fibonacci series using function

```
#include <stdio.h>

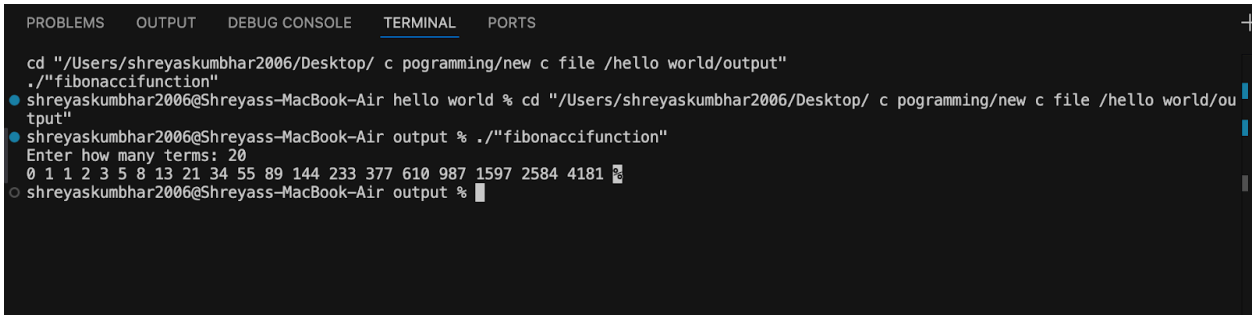
void fibonacci(int n) {
   int a = 0, b = 1, c;

   for(int i = 1; i <= n; i++) {
      printf("%d ", a);
      c = a + b;
      a = b;
      b = c;
   }
}

int main() {
   int n;
   printf("Enter how many terms: ");
   scanf("%d", &n);

   fibonacci(n);

   return 0;
}
```



25)Factorial of a number using functions

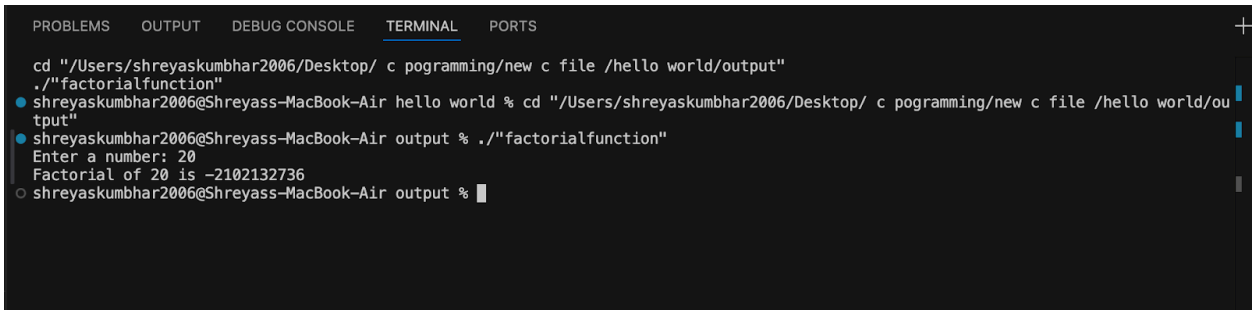
```
#include <stdio.h>

int factorial(int n) {
   int f = 1;
   for(int i = 1; i <= n; i++) {
      f = f * i;
   }
   return f;
}

int main() {
   int n;
   printf("Enter a number: ");
   scanf("%d", &n);

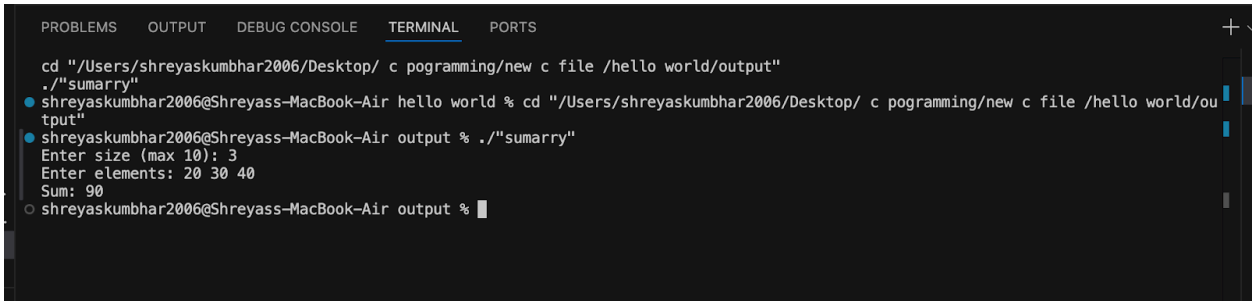
   int result = factorial(n);
   printf("Factorial of %d is %d\n", n, result);

   return 0;
}
```



26) Sum of each element in array

```
#include <stdio.h>
int main() {
   int a[10], n, s = 0, i;
   printf("Enter size (max 10): ");
   if (scanf("%d", &n) != 1 || n < 1 || n > 10) return 1;
   printf("Enter elements: ");
   for (i = 0; i < n; i++) {
      if (scanf("%d", &a[i]) != 1) return 1;
      s = s + a[i];
   }
   printf("Sum: %d\n", s);
   return 0;
}
```

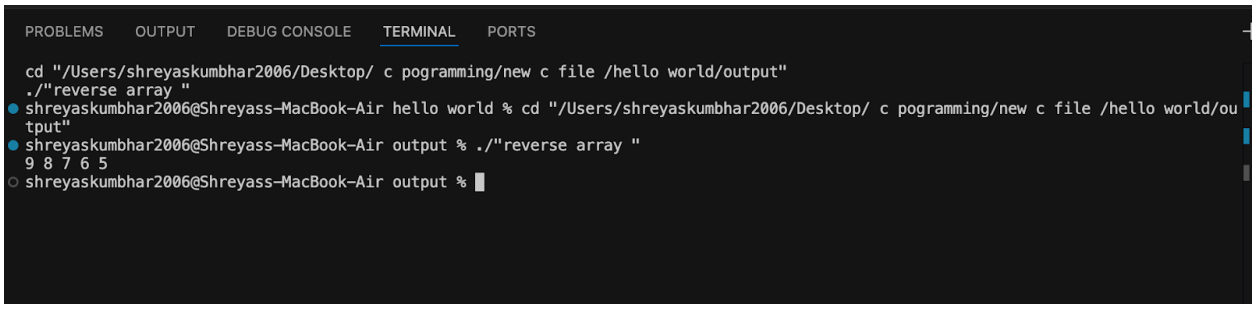


27) Write A c program reversing of an array

```
#include <stdio.h>

void reverse(int a[], int n) {
   int i, t;
   for (i = 0; i < n / 2; i++) {
      t = a[i];
      a[i] = a[n - 1 - i];
      a[n - 1 - i] = t;
   }
}

int main() {
   int a[] = {5, 6, 7, 8, 9};
   int n = 5;
   int i;
   reverse(a, n);
   for (i = 0; i < n; i++) {
      printf("%d ", a[i]);
   }
   printf("\n");
   return 0;
}
```



28) Write a c program to add two matrices

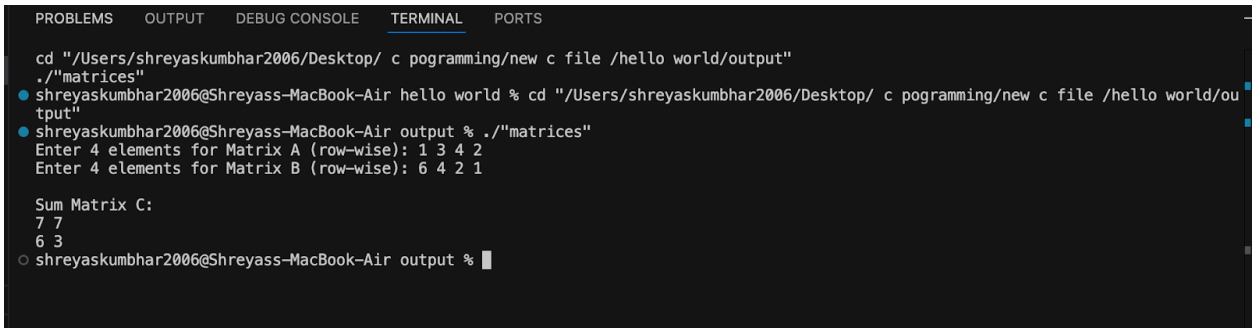
```
#include <stdio.h>
int main() {
   int a[2][2], b[2][2], c[2][2], i, j;

   printf("Enter 4 elements for Matrix A (row-wise): ");
   for (i = 0; i < 2; i++) {
      for (j = 0; j < 2; j++) {
         scanf("%d", &a[i][j]);
      }
   }

   printf("Enter 4 elements for Matrix B (row-wise): ");
   for (i = 0; i < 2; i++) {
      for (j = 0; j < 2; j++) {
         scanf("%d", &b[i][j]);
      }
   }

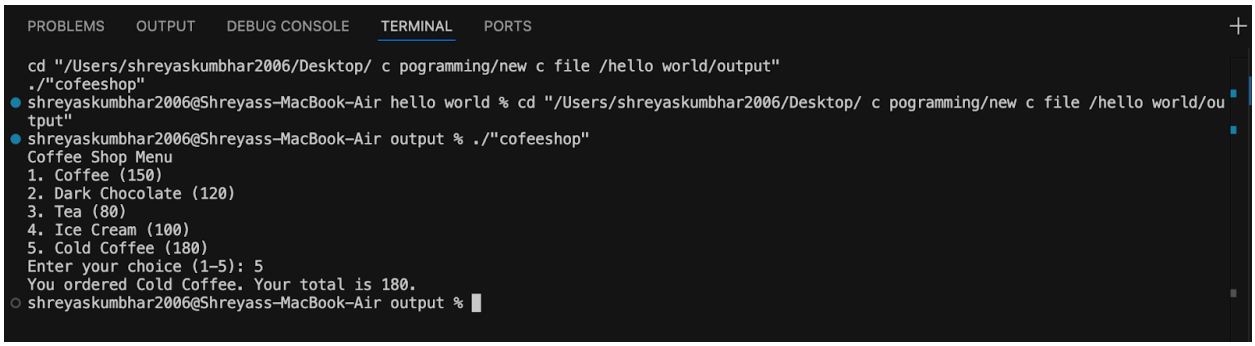
   // Matrix Addition and printing
   printf("\nSum Matrix C:\n");
   for (i = 0; i < 2; i++) {
      for (j = 0; j < 2; j++) {
         c[i][j] = a[i][j] + b[i][j];
         printf("%d ", c[i][j]);
      }
      printf("\n"); // Newline after each row
   }

   return 0;
}
```



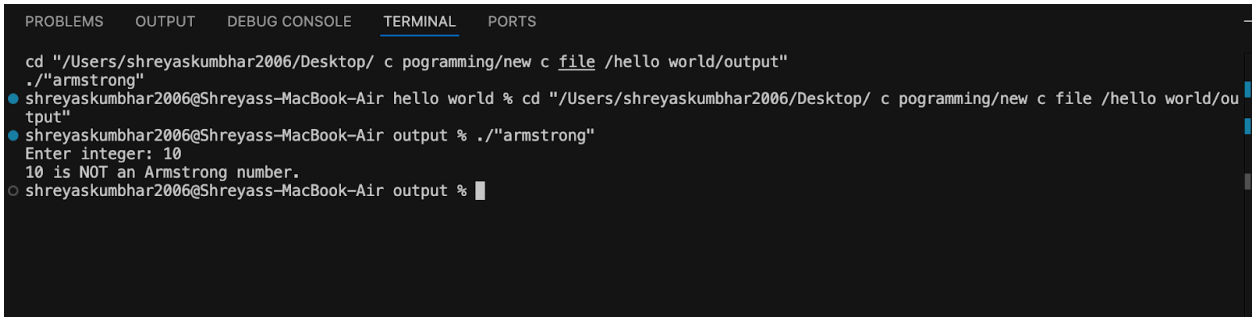
29) Coffee Shop Menu using Switch Case

```
#include <stdio.h>
int main() {
 int choice;
 printf("Coffee Shop Menu \n");
 printf("1. Coffee (150)\n");
 printf("2. Dark Chocolate (120)\n");
 printf("3. Tea (80)\n");
 printf("4. Ice Cream (100)\n");
 printf("5. Cold Coffee (180)\n");
 printf("Enter your choice (1-5): ");
 if (scanf("%d", &choice) != 1) return 1;
 switch (choice) {
  case 1: printf("You ordered Coffee. Your total is 150.\n"); break;
  case 2: printf("You ordered Dark Chocolate. Your total is 120.\n"); break;
  case 3: printf("You ordered Tea. Your total is 80.\n");
  break;
  case 4: printf("You ordered Ice Cream. Your total is 100.\n"); break;
  case 5: printf("You ordered Cold Coffee. Your total is 180.\n"); break;
 default: printf("Invalid choice. Please select a valid number between 1 and 5.\n"); return 0;
 }
}
```



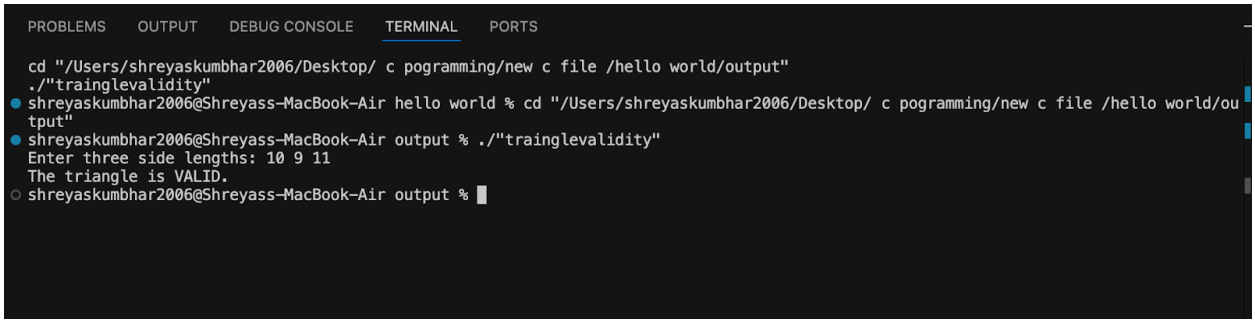
30) Check if a number is Armstrong or Not

```
#include <stdio.h>
int main() {
 int num, originalNum, remainder, result = 0, n = 0, k, p;
 printf("Enter integer: ");
 if (scanf("%d", &num) != 1 || num <= 0) return 1;
 originalNum = num;
 while (num != 0) {
  num /= 10; ++n;
 }
 num = originalNum;
 while (num != 0) {
  remainder = num % 10;
  p = 1;
  for (k = 0; k < n; k++) p *= remainder;
  result += p;
  num /= 10;
 }
 printf("%d is %s.\n", originalNum, (result == originalNum) ? "an Armstrong number" : "NOT an Armstrong
number"); return 0;
}
```



31) Check Triangle Validity

```
#include <stdio.h>
int main() {
 float a, b, c;
 printf("Enter three side lengths: ");
 if (scanf("%f %f %f", &a, &b, &c) != 3 || a <= 0 || b <= 0 || c <= 0) return 1;
 if ((a + b > c) && (a + c > b) && (b + c > a)) {
  printf("The triangle is VALID.\n");
 } else {
  printf("The triangle is NOT valid.\n");
 }
 return 0;
}
```



32) Check if a number is Positive, Negative, or Zero

```
#include <stdio.h>
int main() {
 int num;
 printf("Enter an integer: ");
 if (scanf("%d", &num) != 1) return 1;
 printf("%d is %s.\n", num, (num > 0) ? "Positive" : (num < 0) ? "Negative" : "Zero");
 return 0;
}
```

```
Enter an integer: 20
20 is Positive.


=== Code Execution Successful ===
```

33) Profit and Loss Calculator

```
#include <stdio.h>
int main() {
 float cp, sp;
 printf("Enter Cost Price and Selling Price: ");
 if (scanf("%f %f", &cp, &sp) != 2) return 1;
 if (sp > cp)
 printf("Result: PROFIT of %.2f\n", sp - cp);
 else if (sp < cp)
 printf("Result: LOSS of %.2f\n", cp - sp);
 else
 printf("Result: NO PROFIT, NO LOSS\n");
 return 0;
}
```

```
Enter Cost Price and Selling Price: 50 70
Result: PROFIT of 20.00
```

34) Print Numbers 1 to 100 using For-Loop

```
#include <stdio.h>
int main() {
 int i;
 for (i = 1; i <= 100; i++) {
  printf("%d ", i);
 }
 printf("\n");
 return 0;
}
```

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27
    28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
    51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73
    74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96
    97 98 99 100
```